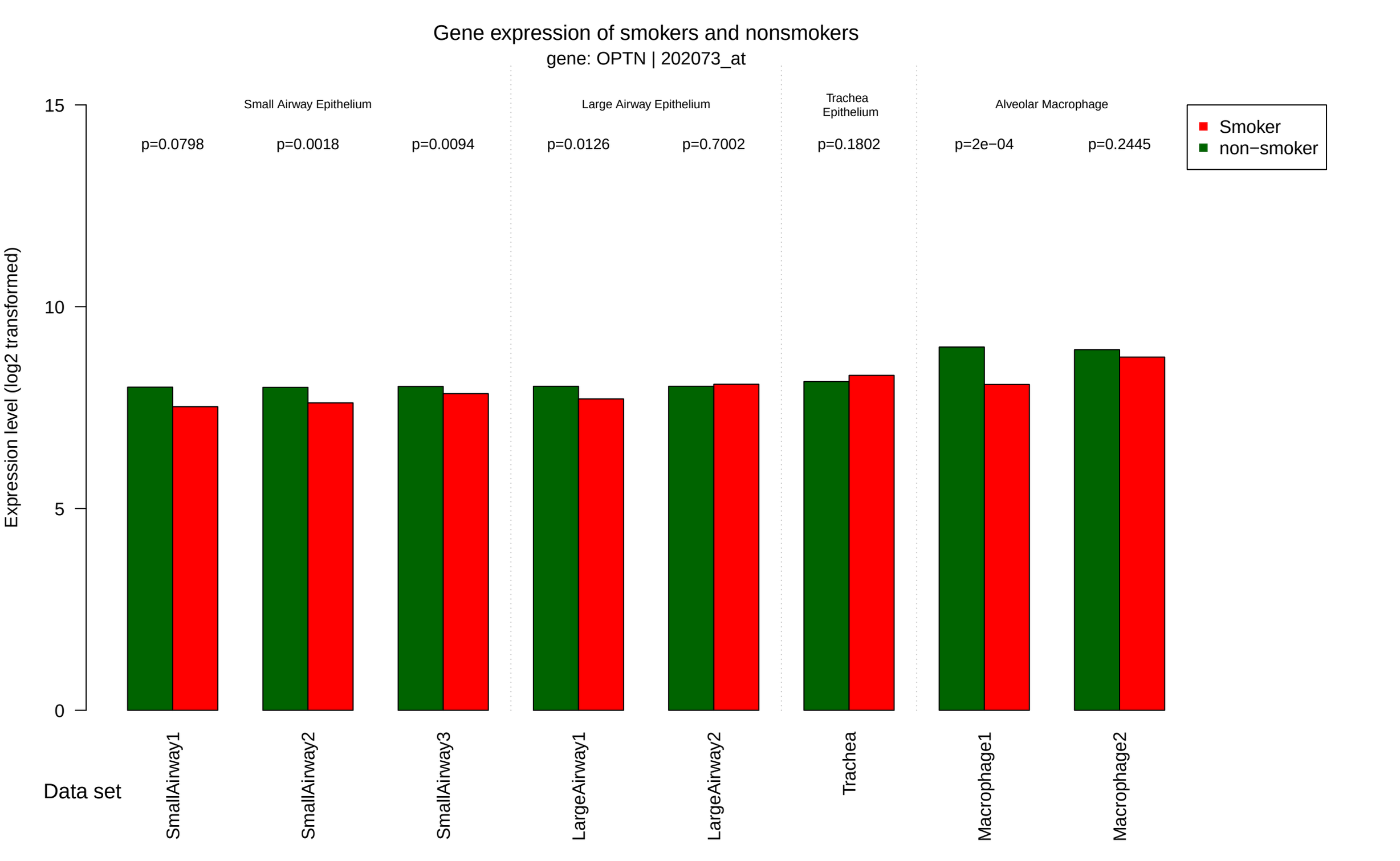
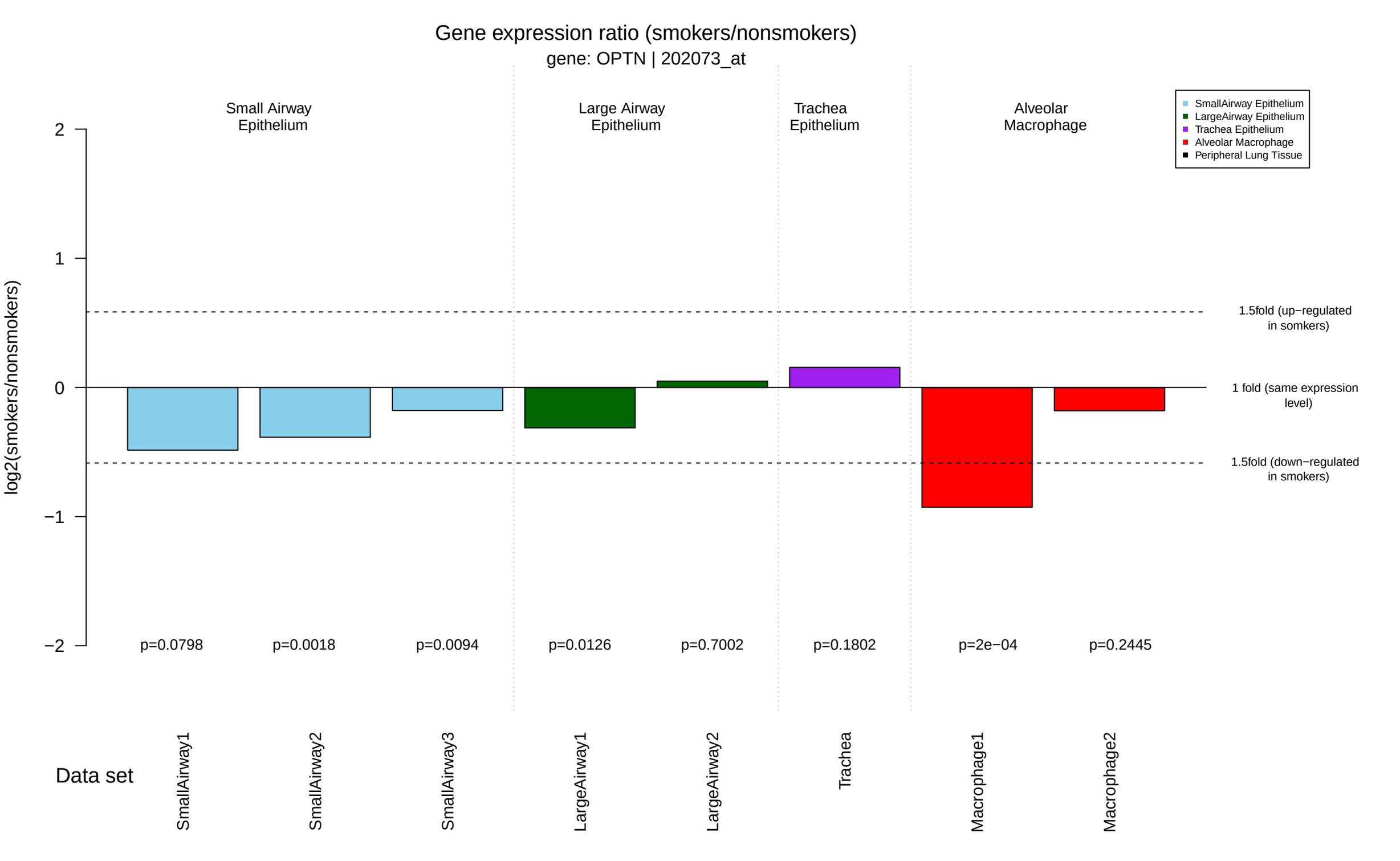
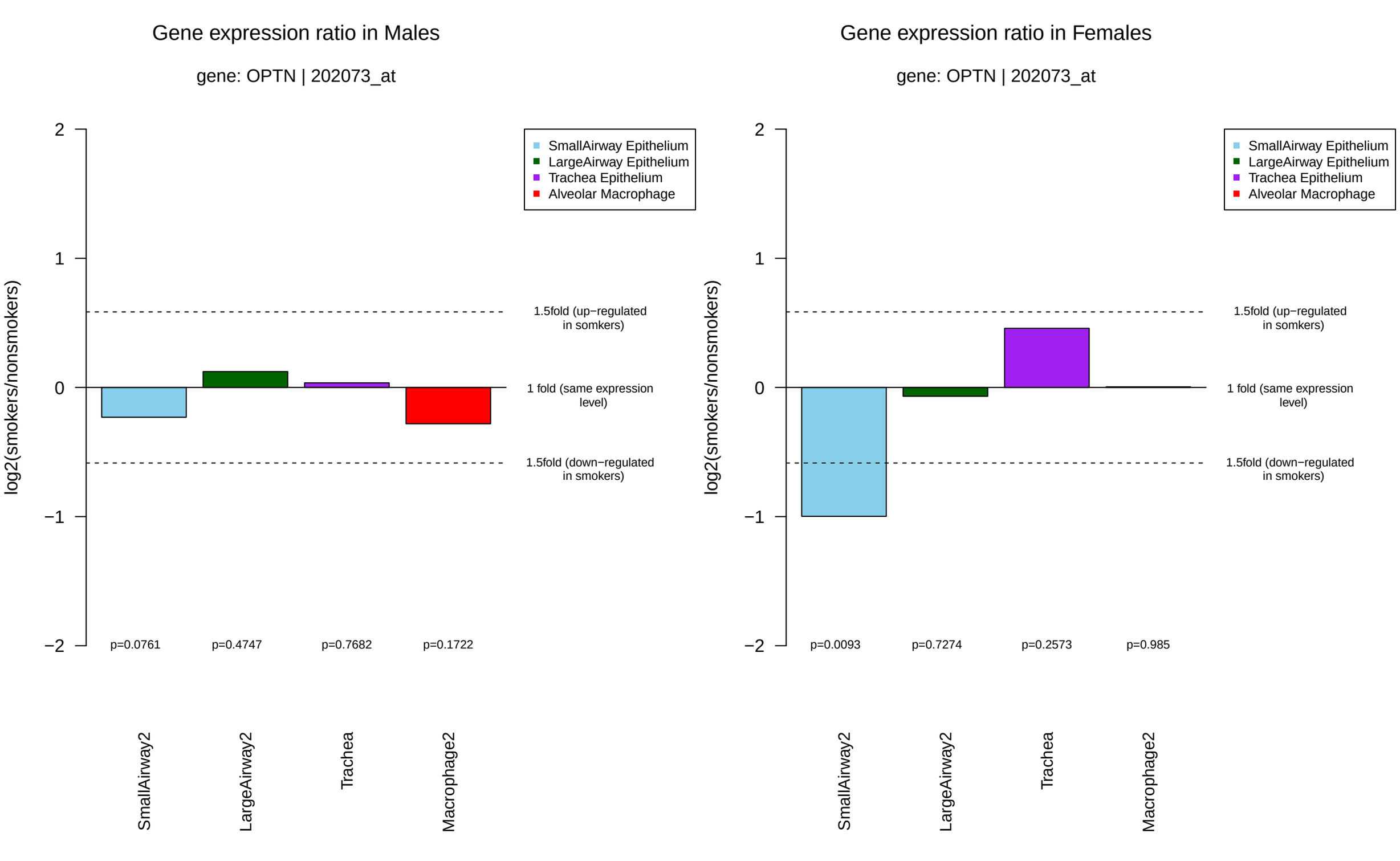
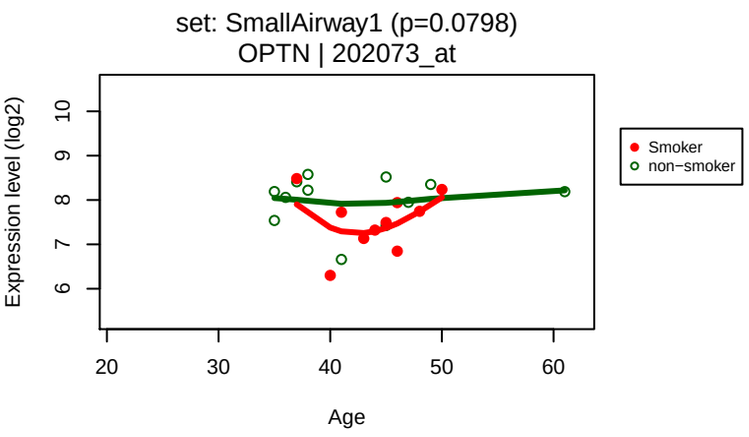
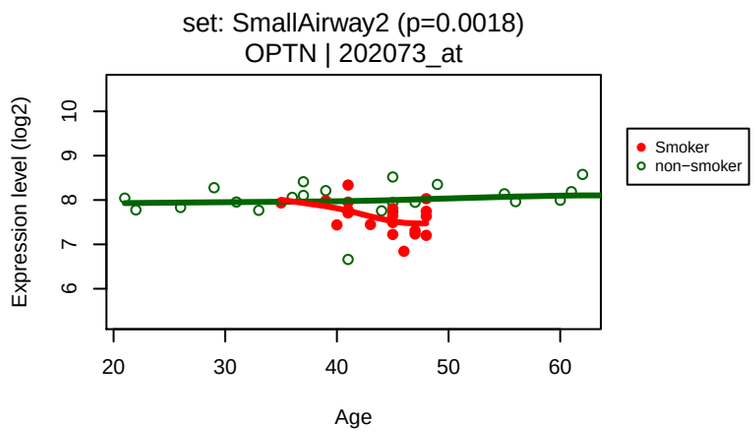
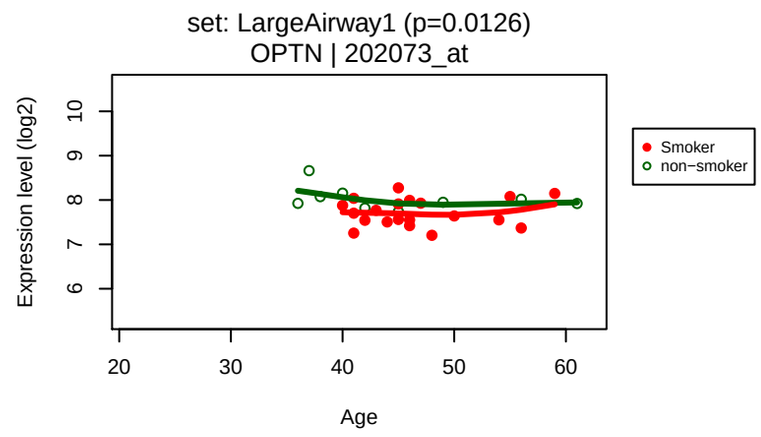

Gene expression of smokers and nonsmokers
gene: OPTN | 202073_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0798
p=0.0018
p=0.0094
p=0.0126
p=0.7002
p=0.1802
p=2e−04
p=0.2445
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

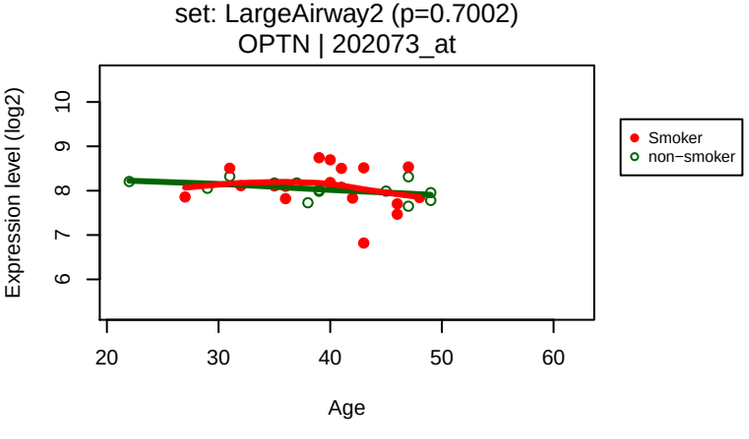

Gene expression ratio (smokers/nonsmokers)
gene: OPTN | 202073_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0798
p=0.0018
p=0.0094
p=0.0126
p=0.7002
p=0.1802
p=2e−04
p=0.2445
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

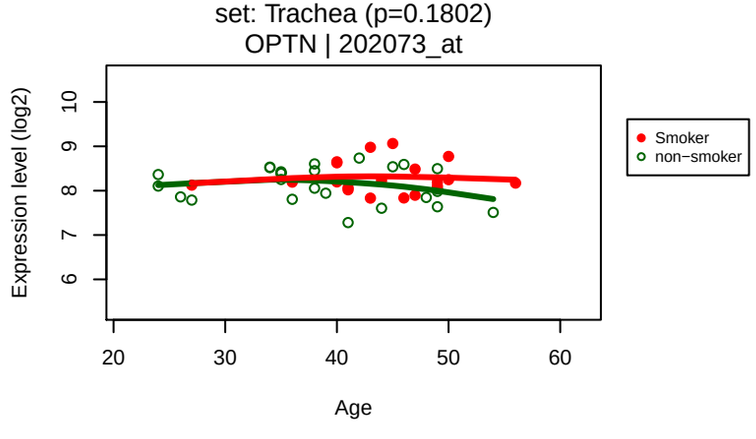

Gene expression ratio in Males
gene: OPTN | 202073_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0761
p=0.4747
p=0.7682
p=0.1722
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPTN | 202073_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0093
p=0.7274
p=0.2573
p=0.985
SmallAirway2
LargeAirway2
Trachea
Macrophage2

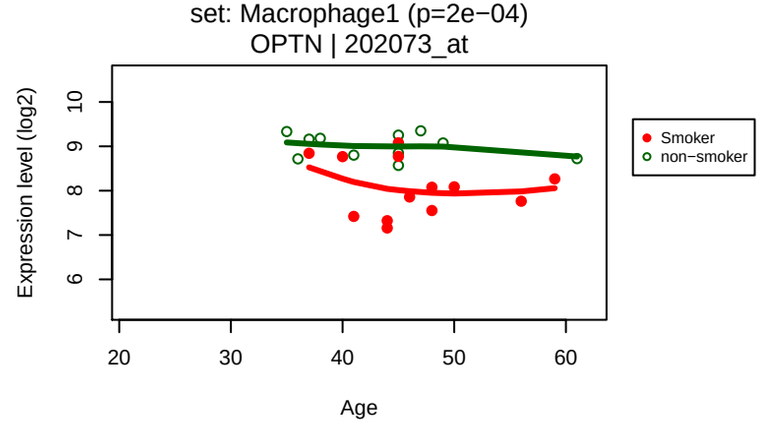

set: SmallAirway1 (p=0.0798)
OPTN | 202073_at
Expression level (log2)
Age
Smoker
non−smoker

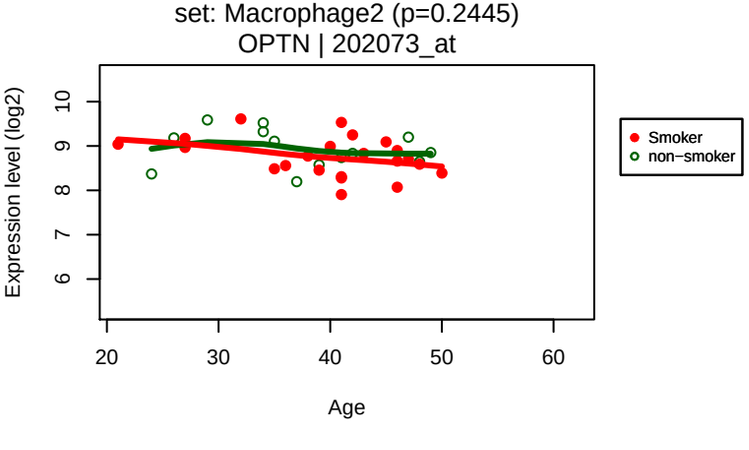

set: SmallAirway2 (p=0.0018)
OPTN | 202073_at
Expression level (log2)
Age
Smoker
non−smoker

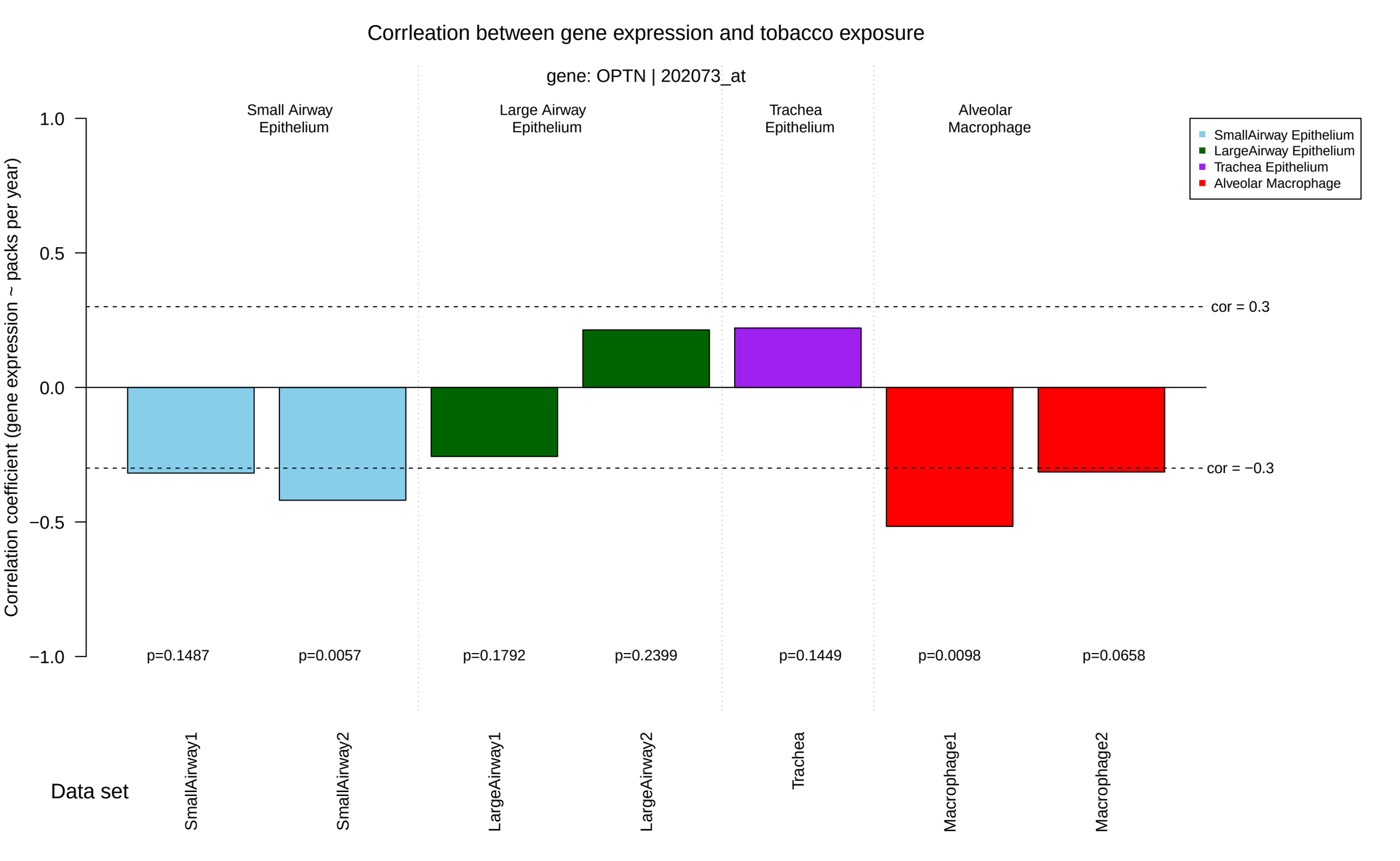

set: LargeAirway1 (p=0.0126)
OPTN | 202073_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7002)
OPTN | 202073_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1802)
OPTN | 202073_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=2e−04)
OPTN | 202073_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2445)
OPTN | 202073_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPTN | 202073_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1487
p=0.0057
p=0.1792
p=0.2399
p=0.1449
p=0.0098
p=0.0658
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2